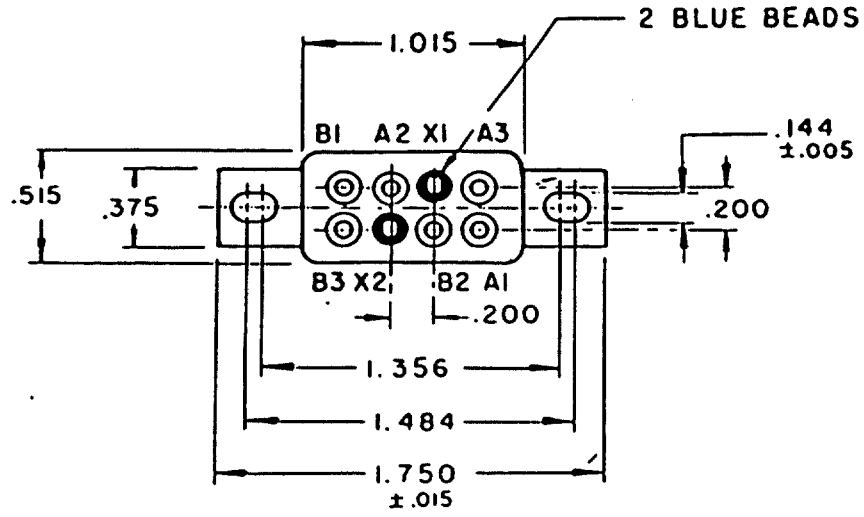
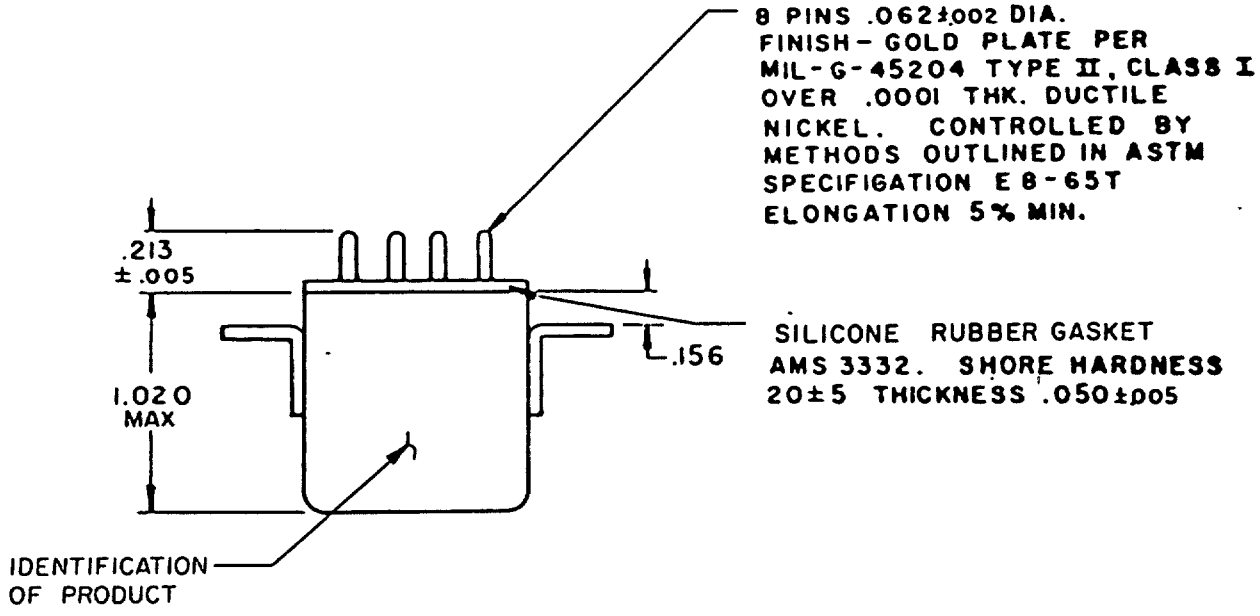
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DASH NUMBERS 1, 2, AND 3

(K) denotes changes

APPROVED 10 Aug 1964 REVISED (J) 5 JUN 87 (K) 10 FEB 89

P.A USAF - 85 Other Cust Navy - EC	International Interest	TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED	MILITARY STANDARD MS27247
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 6

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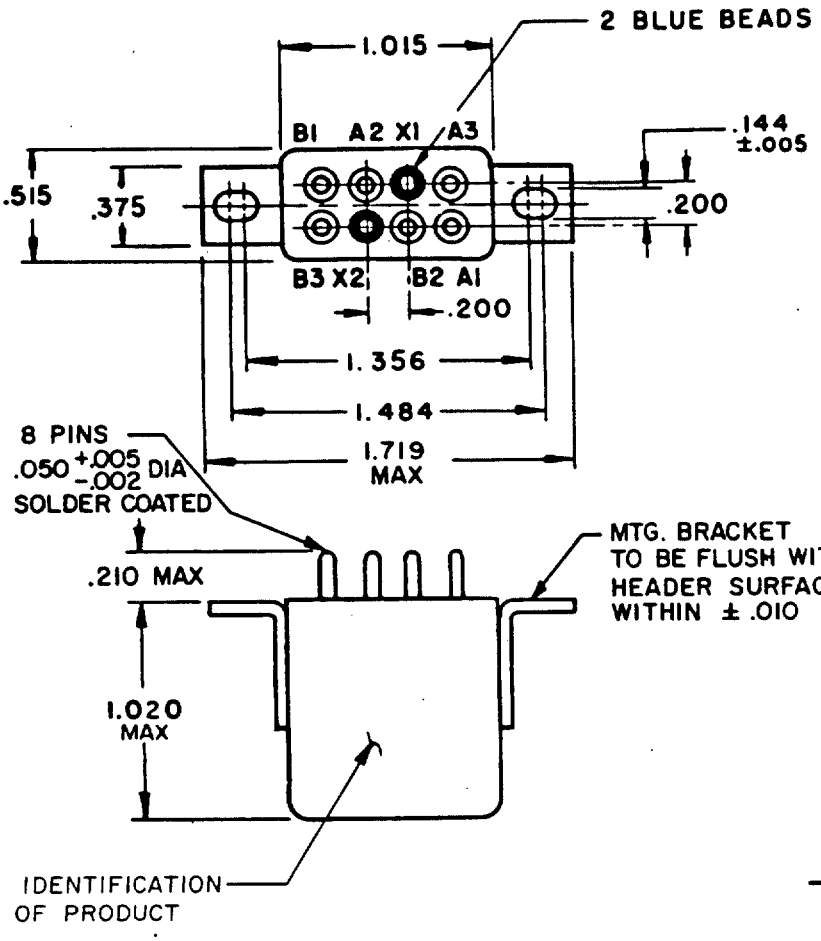
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User activities: Army - Navy - Air Force -

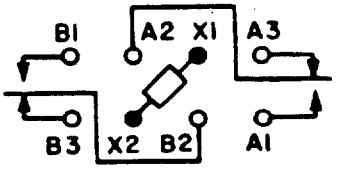
Review activities: Army - AS Navy - Air Force - 11, 99

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DASH NUMBERS 4 AND 5



CIRCUIT DIAGRAM

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P.A USAF - 85 Other Cust Navy - EC	International Interest	TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED	MILITARY STANDARD MS27247
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Inches	mm	Inches	mm
.0001	0.025	.210	5.33
.002	0.05	.213	5.41
.005	0.13	.375	9.52
.010	0.25	.515	13.08
.015	0.38	1.000	25.40
.050	1.27	1.015	25.78
.062	1.57	1.356	34.44
.144	3.66	1.484	37.69
.156	3.96	1.719	43.66
.200	5.08	1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS27247-	Type	Coil	Terminal type	115/200 V ac 3 phase contact rating	Mounting or mating socket	Max weight in pounds
1	I	dc	Plug-in	N/A	N/A	.08
2	I	dc	Plug-in	Applicable	N/A	.08
3	IER	dc	Plug-in	Applicable	N/A	.08
4	I	dc	Printed wire (PW)	N/A	N/A	.08
5	I	dc	Printed wire (PW)	Applicable	N/A	.08

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Review activities: Army - Navy - AS Air Force - 11, 99
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P.A USAF - 85
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International interest
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Review activities: Army -
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SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

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(K) TABLE II. Operating characteristics.

MS part no.	Coil data											Time (milliseconds-maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce			
														Main		Aux	
		Volts 1/	Freq. Hz	Res Ω ±10%	Volts	Amperes	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
MS27247-1	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-2	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-3	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-4	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-5	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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FED. SUP CLASS 5945

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Review activities: Army - Navy - AS Air Force - 11, 99

User activities: Army - Navy - Air Force -

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3 2/	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 2/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10				10				
Inductive	100	4	4			4				4				
Inductive 1/ (-5 only)														
Motor	100	3	3			3				3				
Lamp	100	2	2			2				2				
Transfer load														3/
Mechanical life reduced current	400	2.5	2.5											
Intmd current														4/

1/ Addition rating for -5 only: 2 ampere inductive at 460 V ac, 400 Hz.
2/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3 phase applications.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.
4/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specification except monitored to 3 ohm contact resistance.

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FED. SUP CLASS 5945

P.A USAF - 85 Other Cust Navy - EC

International Interest

Procurement Specification MIL-R-6106

SUPERSEDES:

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Review activities: Army -
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SUPERSEDES:

TITLE

RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

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FED. SUP. CLASS 5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	20 G
Frequency range	10 - 2,000 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,250 V rms	1,000 V rms

Dielectric strength (altitude)

	80,000 ft
Coil to case	350 V rms
Aux contacts	
All other points	350 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	20 amperes, 28 V dc
Rupture current	40 amperes, 115 V ac, 400 Hz
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests are not applicable.

(K) Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military standard sheet.

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